

Technology Stack

Date	06 July 2026
Team ID	
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>Streamlit</i>	<i>Provides interactive UI, waveform visualization, and real-time feedback display</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python</i>	<i>Handles AI processing, speech analysis, and system logic efficiently</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>Local Storage / File System</i>	<i>Stores audio files and generated reports for analysis and download</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Streamlit Cloud</i>	<i>Enables easy deployment and access via web browser</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Supports code management, version tracking, and collaboration</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>OpenAI Whisper, Sentence- BERT, Librosa, Transformer s, Matplotlib, NLTK, Torch</i>	<i>Used for speech-to-text, semantic analysis, audio processing, visualization, and AI-based evaluation</i>